

22/04/2025



Aakash

Medical | IIT-JEE | Foundations

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CODE-A



AIM - 720

(Advanced **INTENSIVE** Mastery for 720)

MM : 720

CST-9

Time : 180 Mins.

Answers

1. (1)	37. (2)	73. (3)	109. (1)	145. (1)
2. (1)	38. (3)	74. (1)	110. (3)	146. (4)
3. (4)	39. (1)	75. (2)	111. (1)	147. (2)
4. (2)	40. (3)	76. (1)	112. (3)	148. (3)
5. (3)	41. (3)	77. (3)	113. (4)	149. (3)
6. (1)	42. (1)	78. (2)	114. (1)	150. (2)
7. (2)	43. (2)	79. (1)	115. (3)	151. (3)
8. (2)	44. (3)	80. (3)	116. (1)	152. (3)
9. (2)	45. (3)	81. (2)	117. (3)	153. (4)
10. (2)	46. (4)	82. (4)	118. (3)	154. (3)
11. (4)	47. (4)	83. (1)	119. (2)	155. (3)
12. (3)	48. (1)	84. (1)	120. (3)	156. (4)
13. (2)	49. (2)	85. (3)	121. (1)	157. (3)
14. (3)	50. (3)	86. (3)	122. (3)	158. (3)
15. (2)	51. (1)	87. (4)	123. (4)	159. (4)
16. (4)	52. (1)	88. (1)	124. (2)	160. (3)
17. (4)	53. (2)	89. (4)	125. (1)	161. (3)
18. (3)	54. (1)	90. (1)	126. (4)	162. (2)
19. (4)	55. (4)	91. (2)	127. (3)	163. (2)
20. (2)	56. (4)	92. (3)	128. (4)	164. (2)
21. (2)	57. (1)	93. (2)	129. (4)	165. (1)
22. (3)	58. (3)	94. (2)	130. (2)	166. (2)
23. (4)	59. (3)	95. (3)	131. (2)	167. (2)
24. (2)	60. (1)	96. (3)	132. (1)	168. (2)
25. (3)	61. (1)	97. (3)	133. (2)	169. (3)
26. (2)	62. (1)	98. (4)	134. (1)	170. (1)
27. (3)	63. (2)	99. (1)	135. (4)	171. (2)
28. (3)	64. (3)	100. (1)	136. (4)	172. (3)
29. (3)	65. (2)	101. (1)	137. (3)	173. (3)
30. (1)	66. (1)	102. (1)	138. (1)	174. (3)
31. (3)	67. (3)	103. (3)	139. (4)	175. (2)
32. (2)	68. (2)	104. (2)	140. (1)	176. (1)
33. (2)	69. (1)	105. (3)	141. (4)	177. (3)
34. (1)	70. (2)	106. (4)	142. (4)	178. (4)
35. (1)	71. (3)	107. (4)	143. (4)	179. (3)
36. (4)	72. (4)	108. (2)	144. (1)	180. (1)

Hints and Solutions

PHYSICS

(1) Answer : (1)

Solution:

Drawing F.B.D. of sphere,



$$\Rightarrow F_B - mg - T = ma$$

$$\Rightarrow d \times \frac{m}{\rho} \times (g + a) - mg - T = ma$$

$$d \times \frac{m}{\rho} (g + a) - mg - ma = T$$

$$T = m(g + a) \left(\frac{d}{\rho} - 1 \right)$$

(2) Answer : (1)

Solution:

$$\text{K.E./Volume} = \frac{1}{2} \rho v^2 = \frac{1}{2} \times 10^3 \times 5 \times 5$$

$$= \frac{25 \times 10 \times 100}{2}$$

$$= 12500 \text{ J/m}^3$$

(3) Answer : (4)

Solution:

Rotational kinetic energy of monoatomic gas is zero. According to law of equipartition of energy, energy is equally divided between each degree of freedom.

(4) Answer : (2)

Solution:

$$\gamma = \frac{C_P}{C_V} \text{ and } C_P - C_V = R$$

$$\Rightarrow C_P - \frac{C_P}{\gamma} = R$$

$$C_P \left[\frac{\gamma - 1}{\gamma} \right] = R$$

$$\Rightarrow C_P = \frac{R\gamma}{\gamma - 1}$$

(5) Answer : (3)

Solution:

A Carnot cycle consists of two adiabatic and two isothermal processes.

Four-steps involved are

(i) Isothermal expansion

(ii) Adiabatic expansion

(iii) Isothermal compression

(iv) Adiabatic compression

(6) Answer : (1)

Solution:

Use, Newton's law of cooling

$$\Rightarrow \frac{60 - 50}{4} \propto [55 - 10]$$

$$\frac{50 - T}{4} \propto \left[25 + \frac{T}{2} - 10 \right]$$

$$\Rightarrow \frac{10}{50 - T} = \frac{45}{15 + \frac{T}{2}}$$

$$\begin{aligned} \Rightarrow 150 + 5T &= 45 \times 50 - 45T \\ 50T &= 45 \times 50 - 50 \times 3 \\ 50T &= 50 (42) \\ T &= 42^\circ\text{C} \end{aligned}$$

(7) Answer : (2)

Solution:

Given $\vec{E} = \alpha x \hat{i}$

(A) $\phi = \int \vec{E} \cdot d\vec{A}$

$$\phi = \int \alpha x \hat{i} \cdot dA \hat{j} = 0$$

(B) $\phi = \int \vec{E} \cdot d\vec{A}, x = a$

$$\phi = \int \alpha x \hat{i} \cdot dA \hat{i} = \int \alpha x dA$$

$$\phi = \alpha a \times a = \alpha a^2$$

(C) $\phi_{\text{net}} = \oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{in}}}{\epsilon_0}$

$$\phi_{\text{enter}} = 0, (\because x = 0)$$

$$\phi_{\text{exit}} = \int \alpha x \hat{i} \cdot dA \hat{i} = \alpha a^3$$

$$\phi_{\text{net}} = \frac{q_{\text{in}}}{\epsilon_0} \Rightarrow \alpha a^3 = \frac{q_{\text{in}}}{\epsilon_0} \Rightarrow q_{\text{in}} = \alpha a^3 \epsilon_0$$

(D) $\phi_{\text{net}} = \oint \vec{E} \cdot d\vec{A}$

Here, $\phi_{\text{net}} = \int \alpha x \hat{i} \cdot dA \hat{i}$

$$\phi_{\text{net}} = \alpha \times (2a) \times (2a)^2 = 8\alpha a^3$$

(8) Answer : (2)

Solution:

$|q| \propto$ number of electric field lines

$$\Rightarrow \frac{|q_1|}{|q_2|} = \frac{12}{8}$$

$$\therefore |q_1| > |q_2|$$

(9) Answer : (2)

Solution:

Potential energy (U) = $\frac{kq_1q_2}{r}$

$$U_i = \frac{k \times 1 \times 2}{1} = 2K \dots (i)$$

$$U_f = 0 \dots (ii)$$

$$W_{\text{ext}} = \Delta U \Rightarrow W_{\text{ext}} = U_f - U_i$$

$$W_{\text{ext}} = -2K$$

(10) Answer : (2)

Solution:

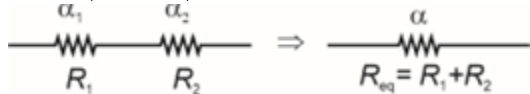
$$P = Vi \Rightarrow 110 = 220 \times i \Rightarrow i = \frac{1}{2} \text{A}$$

$$q = ne \Rightarrow n = \frac{1}{2} \times \frac{1}{1.6 \times 10^{-19}} = 0.3125 \times 10^{19}$$

(11) Answer : (4)

Solution:

$$R = R_0(1 + \alpha \Delta T)$$



$$\begin{aligned}
 R_{eq} &= R_1 + R_2 \Rightarrow (R_1 + R_2)(1 + \alpha \Delta T) \\
 &= R_1(1 + \alpha_1 \Delta T) + R_2(1 + \alpha_2 \Delta T) \\
 (R_1 + R_2)\alpha \Delta T &= R_1 \alpha_1 \Delta T + R_2 \alpha_2 \Delta T \\
 \alpha &= \frac{R_1 \alpha_1 + R_2 \alpha_2}{R_1 + R_2} \\
 \text{Given, } R_1 &= R_2 \\
 \therefore \alpha &= \frac{\alpha_1 + \alpha_2}{2}
 \end{aligned}$$

(12) Answer : (3)

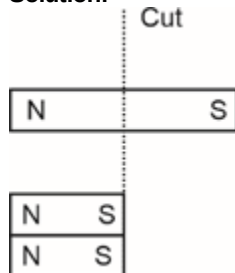
Solution:

For minimum capacitance, series combination is required.

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \Rightarrow C_{eq} = \frac{3}{3} = 1 \mu F$$

(13) Answer : (2)

Solution:



$$M' = \frac{M}{2}$$

$$M_{\text{net}} = \frac{M}{2} + \frac{M}{2} = M$$

(14) Answer : (3)

Solution:

$$\vec{F}_{\text{net}} = q\vec{E} + q(\vec{v} \times \vec{B})$$

(15) Answer : (2)

Solution:

$$\vec{B} = \vec{B}_1 + \vec{B}_2$$

$$B = \frac{\mu_0 I_1}{2\pi R} - \frac{\mu_0 I_2}{2\pi R}$$

$$= \frac{\mu_0}{4\pi} (4 - 3)$$

$$= \frac{\mu_0}{4\pi}$$

(16) Answer : (4)

Solution:

$$\text{emf} = -L \frac{di}{dt}$$

For 0 to $\frac{T}{4}$, $\frac{di}{dt}$ = negative constant $\Rightarrow$ emf = +ve constant

For $\frac{T}{4}$ to $\frac{T}{2}$, $\frac{di}{dt}$ = zero $\Rightarrow$ emf = 0

For $\frac{T}{2}$ to $\frac{3T}{4}$, $\frac{di}{dt}$ = positive constant $\Rightarrow$ emf = -ve constant

(17) Answer : (4)

Solution:

Eddy current effect is used in induction furnace.

(18) Answer : (3)

Solution:

Inductor voltage leads current by 90° .

Over a full cycle, inductor does not consume energy from the voltage source resulting in zero average power dissipation.

(19) Answer : (4)

Solution:

If resistance and capacitive reactance are only present in A.C. circuit, then current leads the voltage.

(20) Answer : (2)

Solution:

Nuclear forces can be attractive or repulsive depending upon the distance between nucleons and is independent of charge on nucleon.

(21) **Answer :** (2)

Solution:

When electron absorbs 12.09 eV energy, then it is excited to $n = 3$

Therefore, increase in angular momentum

$$\Delta L = \frac{3h}{2\pi} - \frac{h}{2\pi}$$

$$= \frac{h}{\pi}$$

(22) **Answer :** (3)

Solution:

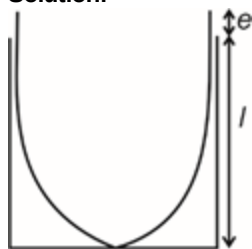
$$h\nu = \phi_0 + \text{K.E.}$$

K.E. depends on incident frequency/wavelength for a given ϕ_0 (Work function depends on material)

K.E. is independent of intensity of incident light.

(23) **Answer :** (4)

Solution:



Wavelength $\lambda = \frac{v}{f} = \frac{340}{340} = 1 \text{ m}$

$$\frac{\lambda}{4} - e = 1$$

$$\Rightarrow e = 25 - 23.5$$

$$= 1.5 \text{ cm}$$

(24) **Answer :** (2)

Solution:

Photo-diode in reverse bias is used to detect intensity of light.

(25) **Answer :** (3)

Solution:

$$(\Delta E_g)_C = 5.5 \text{ eV}$$

$$(\Delta E_g)_{Si} = 1.1 \text{ eV}$$

$$(\Delta E_g)_{Ge} = 0.7 \text{ eV}$$

(26) **Answer :** (2)

Solution:

$$\Delta l = \frac{FL}{AY}$$

$$\frac{5}{\Delta l'} = \frac{\frac{FL}{\pi r^2 Y}}{\frac{FL}{\pi \frac{r^2}{4} Y}}$$

$$\therefore \Delta l' = 20 \text{ mm}$$

(27) **Answer :** (3)

Solution:

$$y = 4 \sin \left(4\pi t - \frac{\pi}{16} x \right)$$

$$\therefore \omega = 4\pi \text{ rad/s, } k = \frac{\pi}{16} \text{ cm}^{-1}$$

$$v = \frac{\omega}{k} = \frac{4\pi}{\frac{\pi}{16}} = 64 \text{ cm/s}$$

$$v = 64 \text{ cm/s in } +x \text{ direction}$$

(28) **Answer :** (3)

Solution:

$$\frac{\text{Kinetic energy}}{\text{Potential energy}} = \frac{\frac{1}{2}m\omega^2(A^2 - x^2)}{\frac{1}{2}m\omega^2 x^2}$$

$$\text{For } x = \frac{A}{2},$$

$$\frac{\text{K.E.}}{\text{P.E.}} = \frac{A^2 - \frac{A^2}{4}}{\frac{A^2}{4}} = \frac{\frac{3}{4}A^2}{\frac{A^2}{4}} = \frac{3}{1}$$

(29) Answer : (3)

Solution:

Energy of EM wave (U) is equally distributed in the form of electric and magnetic field energy.

$$\frac{U_E}{U_B} = 1$$

(30) Answer : (1)

Solution:

$$\text{Fringe width } \beta = \frac{D\lambda}{d}$$

$$(1) \beta' = \frac{\beta}{\mu} \text{ when immersed in liquid } \Rightarrow \beta' < \beta$$

$$(2) \beta \propto \lambda \text{ with increase in } \lambda \Rightarrow \beta \text{ also increases}$$

$$(3) \text{ For white light, bright white fringe is observed at centre.}$$

$$(4) \beta = \frac{(2D)\lambda}{(2)d} \Rightarrow \beta \text{ remains same}$$

(31) Answer : (3)

Solution:

$$\text{By principle of moments, } 2m_1gd_1 = m_2gd_2$$

$$\Rightarrow 2 \times 10 \times 29 = m_2 \times 1$$

$$\Rightarrow m_2 = 580 \text{ g}$$

(32) Answer : (2)

Solution:

Moment of inertia of ring about diameter

$$= \frac{MR^2}{2} = I$$

$$\text{Angular momentum } L = I\omega$$

$$= \frac{MR^2}{2} \omega = 4 \times \frac{(0.25)^2}{2} \times 1000$$

$$= \frac{1000}{8} = 125 \text{ J s}$$

(33) Answer : (2)

Solution:

$$\text{K.E., } K \propto \frac{1}{r} \Rightarrow \frac{K_1}{K_2} = \frac{r_2}{r_1} = \frac{4r}{r}$$

$$= 4 : 1 = \frac{(\text{K.E.})_1}{(\text{K.E.})_2}$$

$$T^2 \propto r^3 \Rightarrow \left(\frac{T_1}{T_2}\right)^2 = \left(\frac{r}{4r}\right)^3 = \frac{1}{64}$$

$$\Rightarrow \frac{T_1}{T_2} = 1 : 8$$

$$L \propto \sqrt{r} \Rightarrow \frac{L_1}{L_2} = \sqrt{\frac{r}{4r}} = 1 : 2$$

$$\text{Total energy } \propto \frac{1}{r} \therefore \frac{TE_1}{TE_2} = \frac{r_2}{r_1} = \frac{4}{1}$$

(34) Answer : (1)

Solution:

$$U_i = -\frac{GMm}{R} \text{ (At the surface)}$$

$$U_f = -\frac{GMm}{2R^3} [3R^2 - r^2]$$

$$\text{At } r = \frac{R}{2}$$

$$U_f = -\frac{GMm}{2R^3} \left[3R^2 - \frac{R^2}{4} \right]$$

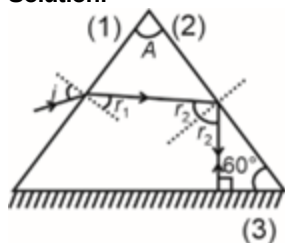
$$= \frac{-GMm1R^2}{8R^3} = -\frac{11}{8} \frac{GMm}{R}$$

$$\text{W.D.} = U_i - U_f = -\frac{GMm}{R} + \frac{11}{8} \frac{GMm}{R}$$

$$= \frac{3}{8} \frac{GMm}{R} = \frac{3}{8} mgR$$

(35) Answer : (1)

Solution:



Ray undergoes TIR at surface (2), as $r_2 = 60^\circ$ and $r_2 > C = \sin^{-1}\left(\frac{1}{\mu}\right) = 45^\circ$

Prism angle $A = r_1 + r_2$

$$\Rightarrow 60^\circ = r_1 + 60^\circ \Rightarrow r_1 = \text{zero}$$

$\therefore i = \text{zero}$

(36) Answer : (4)

Solution:

The angular size of the object equals the angular size of the image in a simple microscope, yet it offers magnification. Thick lenses magnify the effect of dispersion because light travels a longer distance through the lens material, leading to more noticeable colour separation.

(37) Answer : (2)

Solution:

For a number less than 1, all zeroes before non zero number are insignificant.

(38) Answer : (3)

Solution:

Joule is unit of energy

SI	New system
$n_1 = 10$	$n_2 = ?$
$M_1 = 1 \text{ kg}$	$M_2 = \gamma \text{ kg}$
$L_1 = 1 \text{ m}$	$L_2 = \beta \text{ m}$
$T_1 = 1 \text{ s}$	$T_2 = \alpha \text{ s}$

Dimensional formula of energy is $[ML^2 T^{-2}]$

Comparing with $[M^a L^b T^c]$

$$a = 1, b = 2, c = -2$$

$$n_2 = n_1 \left(\frac{M_1}{M_2}\right)^a \left(\frac{L_1}{L_2}\right)^b \left(\frac{T_1}{T_2}\right)^c$$

$$n_2 = 10 \left(\frac{1 \text{ kg}}{\gamma \text{ kg}}\right)^1 \left(\frac{1 \text{ m}}{\beta \text{ m}}\right)^2 \left(\frac{1 \text{ s}}{\alpha \text{ s}}\right)^{-2}$$

$$= \frac{10\alpha^2}{\gamma\beta^2} = 10\alpha^2\gamma^{-1}\beta^{-2}$$

(39) Answer : (1)

Solution:

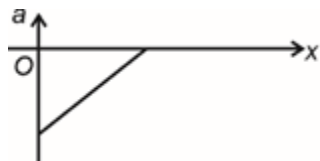
displacement (x) is downwards, $x < 0$

velocity (v) is downwards, $v < 0$

and acceleration is opposite to the direction of velocity, so $a > 0$.

(40) Answer : (3)

Solution:



$$v = -\left(\frac{v_0}{x_0}\right)x + v_0$$

$$a = \frac{v dv}{dx}$$

$$a = \left[-\left(\frac{v_0}{x_0}\right)x + v_0\right] \left[-\frac{v_0}{x_0}\right] \Rightarrow a = \left(\frac{v_0}{x_0}\right)^2 x - \frac{v_0^2}{x_0}$$

(41) Answer : (3)

Solution:



$$\vec{F} = (4\hat{i} + 3\hat{j})\text{ N}$$

$$|\vec{F}| = \sqrt{16+9} = 5\text{ N}$$

$$a = \frac{|\vec{F}_{\text{net}}|}{m} = \frac{5}{10} = \frac{1}{2} = 0.5\text{ m/s}^2$$

$$\vec{a} = \frac{4\hat{i}+3\hat{j}}{10} = (0.4\hat{i} + 0.3\hat{j})\text{ m/s}^2$$

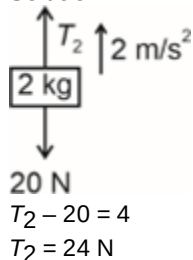
$$\tan \alpha = \frac{3}{4}$$

$$\alpha = \tan^{-1}\left(\frac{3}{4}\right)$$

$$= 37^\circ \text{ w.r.t. } 4\text{ N force}$$

(42) Answer : (1)

Solution:



$$T_2 - 20 = 4$$

$$T_2 = 24\text{ N}$$



$$T_1 - 84 = 12$$

$$T_1 = 84 + 12$$

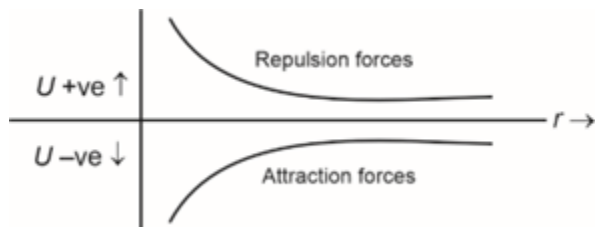
$$= 96\text{ N}$$

(43) Answer : (2)

Solution:

Change in potential energy = -(work done by conservative force)

$$\Rightarrow \Delta U = -W_C$$



(44) Answer : (3)

Solution:

For conservative force $F = -\frac{dU}{dx}$

$$= -(2bx - c)$$

$$= -2bx + c$$

At equilibrium, $F = 0 \Rightarrow -2bx + c = 0$

$$c = 2bx$$

$$x = \frac{c}{2b}$$

(45) Answer : (3)

Solution:

Given, $R = 4H$

$$\frac{u^2 2 \sin \theta \cos \theta}{g} = 4 \frac{u^2 \sin \theta \sin \theta}{2g}$$

$$\tan \theta = 1$$

$$\theta = 45^\circ$$

CHEMISTRY

(46) Answer : (4)

Solution:

a. Chloroform and aniline – Distillation

b. Glycerol from spent-lye – Distillation under reduced pressure

c. Aniline – water mixture – Steam distillation

d. Calcium sulphate and camphor– Sublimation

(47) Answer : (4)

Solution:

$-\text{CH}_3$ group has +I effect while $-\text{Cl}$ has –I effect,

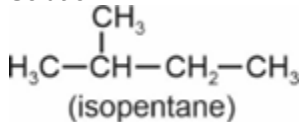
Presence of alkyl group on carboxylic acid decreases its acidic strength by destabilising carboxylate ion formed after loss of proton while an electronegative atom with –I effect increases the acidic strength by stabilising carboxylate ion

Hence, order of acidic strength is

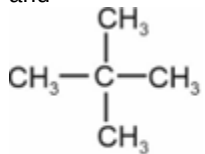


(48) Answer : (1)

Solution:

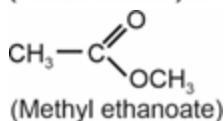
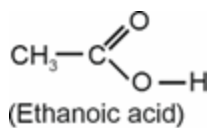


and



(Neopentane)

are chain isomers.



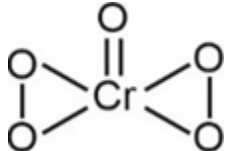
(49) Answer : (2)

Solution:

1. Water gas – $\text{CO} + \text{H}_2$
2. Producer gas – $\text{CO} + \text{N}_2$
3. Cryolite – Na_3AlF_6
4. Borax – $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$

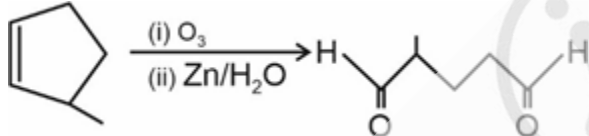
(50) Answer : (3)

Solution:



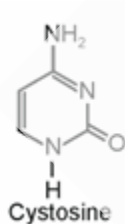
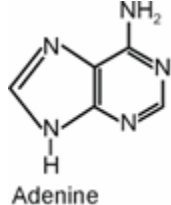
(51) Answer : (1)

Solution:



(52) Answer : (1)

Solution:



(53) Answer : (2)

Solution:

$$r_n \propto \frac{n^2}{Z}$$

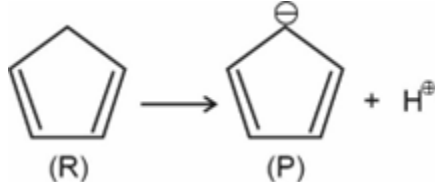
$$\Rightarrow \frac{r_{\text{He}^+}}{r_{\text{H}}} = \frac{n^2(\text{He}^+)}{Z(\text{He}^+)} \times \frac{Z(\text{H})}{n^2(\text{H})}$$

$$\Rightarrow \frac{a}{r_{\text{H}}} = \frac{1}{2} \times \frac{1}{3^2}$$

$$\Rightarrow r_{\text{H}} = 18a \text{ \AA}$$

(54) Answer : (1)

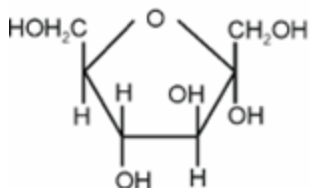
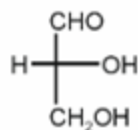
Solution:



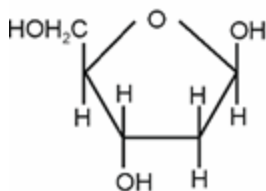
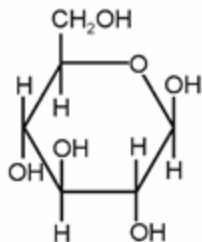
Species (P) is aromatic in nature hence acidic nature of (R) is maximum among the given options.

(55) Answer : (4)

Solution:


 α -D-(-)-Fructofuranose


D-(+)-Glyceraldehyde


 β -D-2-deoxyribose

 β -D-(+)-Glucopyranose

(56) Answer : (4)

Solution:

In photoelectric effect, the number of electrons ejected is proportional to the intensity of the light used.

(57) Answer : (1)

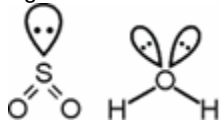
Solution:

The magnitude of H-bonding depends on the physical state of the compound. It is maximum in the solid state and minimum in the gaseous state.

(58) Answer : (3)

Solution:

AgCl is more covalent than NaCl.



$\Rightarrow$ Both are bent

-Cl is more electronegative than I. Hence, HI is less polar than HCl.

(59) Answer : (3)

Solution:

$$\mu_{\text{observed}} = 1 \times 3.33 \times 10^{-30} \text{ Cm}$$

$$\mu_{\text{calculate}} = 1.602 \times 10^{-19} \times 100 \times 10^{-12} \text{ Cm}$$

$$= 1.6 \times 10^{-29} \text{ Cm}$$

$$\text{Percentage ionic character} = \frac{1 \times 3.33 \times 10^{-30} \times 100}{1.6 \times 10^{-29}}$$

$$= 20.81\%$$

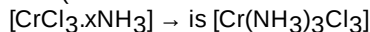
(60) Answer : (1)

Solution:

$$\Delta T_b(\text{complex}) = \Delta T_b(\text{glucose})$$

$$iK_b m = K_b \times m$$

$i = 1$ (So no dissociation or association)



Hence, $x = 3$

(61) Answer : (1)

Solution:

$$i = 1 - \alpha + n\alpha$$

$$= 1 - 0.1 + 2 \times 0.1$$

$$= 1.1$$

$$\Delta T_f = iK_f m$$

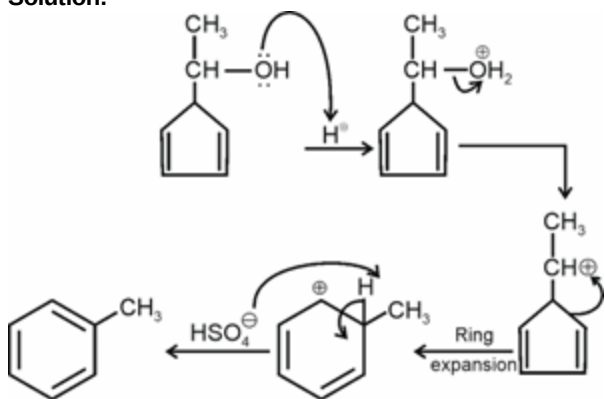
$$= 1.1 \times 1.86 \times 0.1$$

$$= 0.205$$

$$\therefore T_f \approx -0.205$$

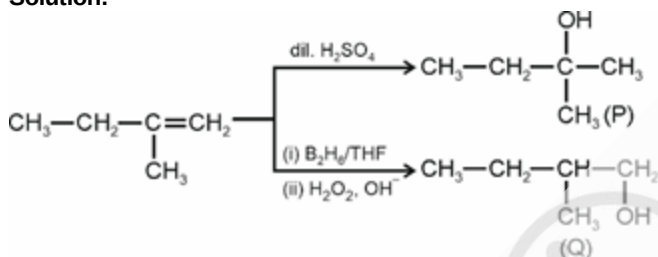
(62) Answer : (1)

Solution:



(63) Answer : (2)

Solution:



(P) and (Q) are tertiary and primary alcohols respectively therefore (P) and (Q) can be distinguished by Lucas reagent.

(64) Answer : (3)

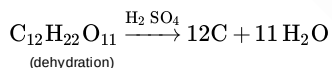
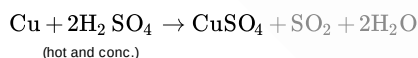
Solution:



Boiling points of alcohols are higher than those of ethers due to the presence of intermolecular hydrogen bonding in alcohols.

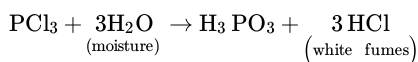
(65) Answer : (2)

Solution:



(66) Answer : (1)

Solution:



(67) Answer : (3)

Solution:

Fluorine is stronger oxidising agent than chlorine
 $\cdot F_2$ has lower bond dissociation enthalpy than Cl_2 .

(68) Answer : (2)

Solution:

For the same amount current passed,

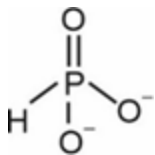
$$\frac{W_{Ag}}{108} = \frac{2800}{22400} \times 2$$

$$W_{Ag} = 27$$

(69) Answer : (1)

Solution:

HPO_3^{2-} does not contain any acidic hydrogen, hence will not act as Bronsted acid.



H_3PO_3 cannot accept H^+ ; hence can not be treated as Bronsted base.

HPO_4^{2-} has one acidic hydrogen and can take H^+ ion also. Hence, it can be act as both Bronsted acid and Bronsted base.

(70) Answer : (2)

Solution:

$$\Delta_m^\circ(\text{CH}_3\text{COONa}) = \Delta_m^\circ(\text{CH}_3\text{COONa}) + \Delta_m^\circ(\text{HCl}) - \Delta_m^\circ(\text{NaCl})$$

(71) Answer : (3)

Solution:

$$[\text{H}^+] = \frac{(0.2 \times 1 \times 1) + (0.1 \times 2 \times 1) + (0.2 \times 1 \times 2)}{1 + 2 + 1}$$

$$[\text{H}^+] = \frac{0.2 + 0.2 + 0.4}{4} = \frac{0.8}{4} = 0.2$$

$$\text{pH} = -\log[\text{H}^+] = -\log(0.2) = 0.69$$

(72) Answer : (4)

Hint:

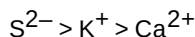
The maximum covalency of B is 4.

In $[\text{AlCl}(\text{H}_2\text{O})_5]^{2+}$, the covalency of Al is 6.

(73) Answer : (3)

Solution:

Cations are smaller and anions are larger as compare to the parent atom.



(74) Answer : (1)

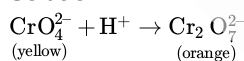
Solution:

Np and Pu are the only actinoids which exhibit +7 oxidation state.

Pr, Nd, Tb and Dy exhibit +4 state but only in oxides (MO_2).

(75) Answer : (2)

Solution:



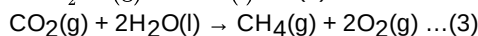
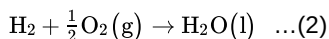
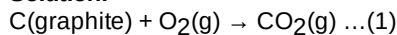
(76) Answer : (1)

Solution:

The third law of thermodynamics is only applicable to pure crystalline substances, as there is perfect order in a crystal at absolute zero.

(77) Answer : (3)

Solution:



$$\text{Eq. (1)} + 2 \times \text{eq. (2)} + \text{eq. (3)}$$

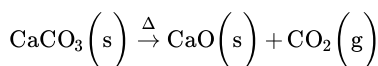
$$\Delta_r H^\circ = (-393.5 + 2(-285.8) + 890.3) \text{ kJ mol}^{-1}$$

$$= -393.5 - 571.6 + 890.3$$

$$= -74.8 \text{ kJ mol}^{-1}$$

(78) Answer : (2)

Solution:



$$1 \text{ mole CaCO}_3 = 1 \text{ mole CaO}$$

$$\text{Mass of CaCO}_3 (95\% \text{ pure}) = 200 \times \frac{95}{100} = 190 \text{ g}$$

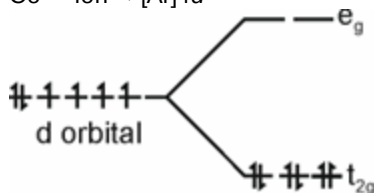
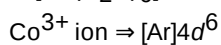
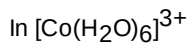
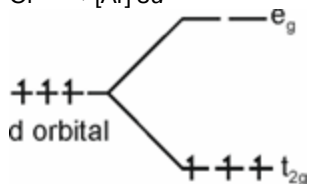
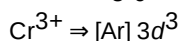
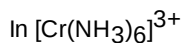
$$100 \text{ g of CaCO}_3 = 56 \text{ g of CaO}$$

$$\therefore 190 \text{ g of CaCO}_3 = \frac{56}{100} \times 190$$

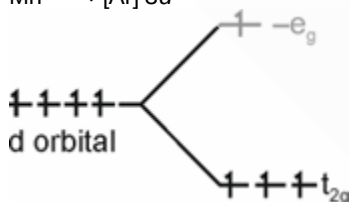
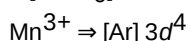
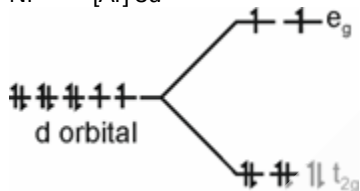
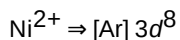
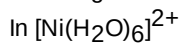
$$= 106.4 \text{ g}$$

(79) Answer : (1)

Solution:



Water ligand with Co^{3+} act as strong field ligand.



(80) Answer : (3)

Solution:

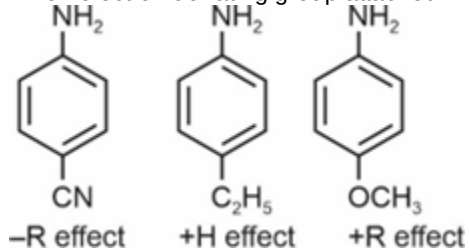
Complexes in which a metal is bound to more than one kind of donor groups.

$[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NO}_2)]$ is known as hetroleptic complex.

(81) Answer : (2)

Solution:

When electron donating group attached with benzene ring it increases the basicity of aniline.

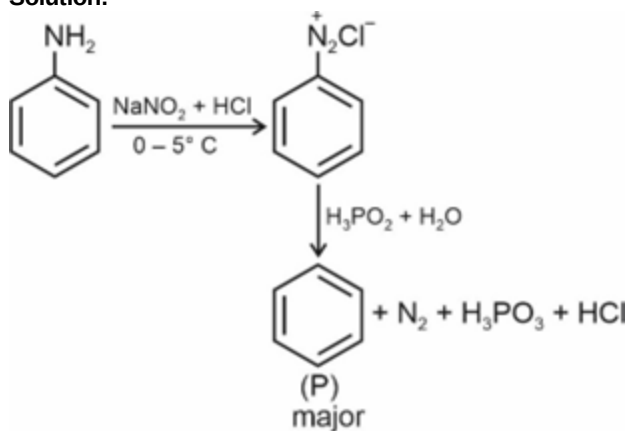


So, correct order of basicity is

(iv) > (iii) > (i) > (ii)

(82) Answer : (4)

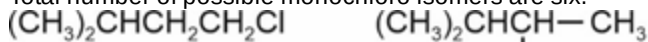
Solution:



(83) Answer : (1)

Solution:

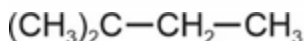
Total number of possible monochloro isomers are six.



(i)

Cl

(ii) optically active



Cl

(iii)

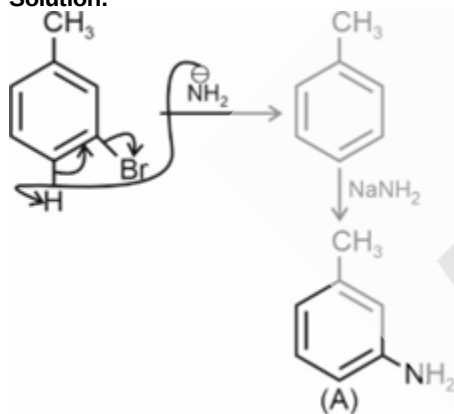


CH₂-Cl

(iv) optically active

(84) Answer : (1)

Solution:



(85) Answer : (3)

Solution:

For zero order reaction

$$t_{100\%} = 2t_{50\%}$$

$$= 2 \times 15 = 30 \text{ min}$$

(86) Answer : (3)

Solution:

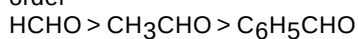
$$\frac{r_2}{r_1} = 2^{\frac{30}{10}} = 8$$

$$r_2 = 8r_1$$

(87) Answer : (4)

Solution:

As the size of alkyl groups increases reactivity at aldehydes towards nucleophilic addition reaction decreases hence correct order



(88) Answer : (1)

Solution:

Aldehydes that do not have any α -H give Cannizzaro reaction.

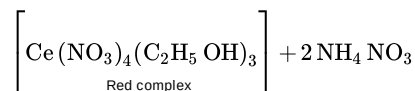
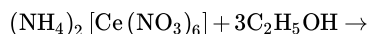
(89) Answer : (4)

Solution:

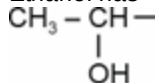
Sodium nitroprusside $\text{Na}_2[\text{Fe}(\text{CN})_6\text{NO}]$ is used to detect sulphide ion.

(90) Answer : (1)

Solution:



Ethanol has



group gives positive iodoform test.

BIOLOGY

(91) Answer : (2)

Solution:

Shoot apical meristem occupies the distant most region of stem axis.

(92) Answer : (3)

Solution:

Pyrimidines are heterocyclic, 6-membered single-ring structure with N at 1 and 3 position, e.g. cytosine, thymine and uracil.

(93) Answer : (2)

Solution:

A new aminoacyl tRNA complex reaches A-site and forms codon-anticodon bonding. This requires elongation factor and GTP. Peptidyl transferase catalyses peptide bond formation.

(94) Answer : (2)

Solution:

Dicot stem	–	Presence of collenchymatous hypodermis and almost uniform sized vascular bundles
Dicot root	–	Presence of diarch to tetrarch vascular bundles, no cuticle is found on epidermis
Monocot stem	–	Presence of scattered vascular bundles but devoid of phloem parenchyma
Monocot leaf	–	Presence of Bulliform cells which help in rolling of leaves during drought

(95) Answer : (3)

Solution:

The three-dimensional structure of tRNA was proposed to be inverted L-shaped (by Kim and Klug). This is the actual structure of tRNA.

(96) Answer : (3)

Solution:

Phloem fibres are sclerenchymatous fibres. They do not have branches but have needle-like apices and are formed during secondary growth. These are used commercially for various purposes.

(97) Answer : (3)

Solution:

RNA polymerase facilitates the opening of DNA double helix during transcription.

(98) Answer : (4)

Solution:

Untranslated regions are required for efficient translation process.

(99) Answer : (1)

Solution:

Metacentric chromosomes are also known as the isobrachial chromosomes.

(100) Answer : (1)

Solution:

Each tetrad has 4 chromatids, so there will be $4 \times 15 = 60$ chromatids in the pachytene stage.

(101) Answer : (1)

Solution:

According to fluid mosaic model given by Nicolson and Singer, the quasi fluid nature of lipid enables lateral movement of protein within the overall bilayer.

(102) Answer : (1)

Solution:

Duplication of semiautonomous organelles is associated with the G₂ phase of cell cycle.

(103) Answer : (3)

Solution:

During meiosis I and mitosis II, nuclear membrane disappears and microtubules attach to the kinetochores of chromosomes.

(104) Answer : (2)

Solution:

In bryophytes, main plant body is gametophyte. Sporophyte is not free living but attached to the photosynthetic gametophyte. Zygotes do not undergo reduction division immediately.

(105) Answer : (3)

Solution:

Brown algae have cell wall covered with algin.
Stored food is mannitol and laminarin in brown algae.

(106) Answer : (4)

Solution:

A nucleotide has three chemically distinct components. One is a heterocyclic nitrogenous base, the second is a monosaccharide and the third is phosphate moiety.

(107) Answer : (4)

Solution:

In the year 1963, the two enzymes responsible for restricting the growth of bacteriophage in *Escherichia coli* were isolated. One of these added methyl groups to DNA, while the other cut DNA. The latter was called restriction endonuclease.

(108) Answer : (2)

Solution:

GLUT-4 enables glucose transport into cells. Ribulose Bisphosphate Carboxylase-Oxygenase (RuBisCO) is the most abundant protein in the whole of the biosphere.

(109) Answer : (1)

Solution:

Pigments	Carotenoids, Anthocyanins, etc.
Alkaloids	Morphine, Codeine, etc.
Terpenoids	Monoterpenes, Diterpenes, etc.
Essential oils	Lemon grass oil, etc.
Toxins	Abrin, Ricin
Lectins	Concanavalin A
Drugs	Vinblastin, Curcumin, etc.
Polymeric Substances	Rubber, Gums, Cellulose

(110) Answer : (3)

Solution:

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Skeletal muscle fibres	Cardiac muscle fibres
Unbranched	Branched
Multinucleated	Uninucleated
Striated	Striated
Communication junctions absent	Communication junctions present

(111) Answer : (1)

Solution:

Golden rice is vitamin 'A' enriched and can prevent problem of night blindness due to vitamin 'A' deficiency.

(112) Answer : (3)

Solution:

Neurons, the unit of neural system, are excitable cells. The neuroglial cells which constitute the rest of the neural system protect and support neurons. Neuroglia make up more than one half the volume of neural tissue in our body. They do not have the property of contractility or extensibility.

(113) Answer : (4)

Solution:

Clavicle and sternum are commonly called collar bone and breast bone respectively.

First cervical and second cervical vertebrae are called atlas and axis respectively.

(114) Answer : (1)

Solution:

People who have Rh antigen on their RBCs are called Rh positive (Rh + ve) and those in whom this antigen is absent are called Rh negative (Rh – ve).

Blood Group	Antigens on RBCs	Antibodies in Plasma	Donor's Group
A	A	anti-B	A, O
B	B	anti-A	B, O
AB	A, B	nil	AB, A, B, O
O	nil	anti-A, B	O

(115) Answer : (3)

Solution:

Neural signals through the sympathetic nerves (part of ANS) can increase the rate of heart beat, the strength of ventricular contraction and thereby cardiac output. On the other hand, parasympathetic neural signals decrease the rate of heart beat, speed of conduction of action potential and thereby the cardiac output. Adrenal medullary hormones can also increase the cardiac output.

(116) Answer : (1)

Solution:

For more than hundred years, microbes are being used to treat sewage by the process of activated sludge formation and thus, help in recycling of water.

(117) Answer : (3)

Solution:

About 99% of animals are conformers.

(118) Answer : (3)

Solution:

Organisms are taken out of their natural habitat in *ex-situ* conservation.

(119) Answer : (2)

Solution:

Passenger pigeons have become extinct in last 500 years due to overexploitation by humans.

(120) Answer : (3)

Solution:

Gametes are non-flagellated and similar in size in the sexual life cycle of *Spirogyra*.

(121) Answer : (1)

Solution:

Major component of biogas is methane, carbon dioxide and mixture of other gases H₂, H₂S, etc.

(122) Answer : (3)

Solution:

In mutualism, both partners are benefitted.

(123) Answer : (4)

Solution:

Joseph Priestley revealed the essential role of air in the growth of green plants.

Jan Ingenhousz showed that it is only green parts of the plant that could release oxygen.

T.W. Engelmann used green alga, *Cladophora*, and a suspension of bacteria in his experiment.

Cornelius van Neil inferred from his experiment that O_2 evolved by the green plants comes from H_2O not from CO_2 .

(124) Answer : (2)

Solution:

Availability of sunlight is an external factor not an internal factor.

(125) Answer : (1)

Solution:

Direct synthesis of ATP in a reaction is known as substrate level phosphorylation. This occurs at two steps of glycolysis and one step of Krebs cycle.

Hence, during complete oxidation of one molecule of glucose, substrate level phosphorylation occurs 6 times. (3 steps)

(126) Answer : (4)

Solution:

Zooplanktons, being consumers, can not occupy first trophic level.

(127) Answer : (3)

Solution:

Citric acid is a six carbon containing compound.

(128) Answer : (4)

Solution:

Pyramid of number of tree ecosystem is spindle shaped.

This ecological pyramid does not represent the pyramid of number for a forest ecosystem.

(129) Answer : (4)

Solution:

Systematics takes into account evolutionary relationships between organisms.

(130) Answer : (2)

Solution:

Ascomycetes is known as sac fungi. *Claviceps* is an example of ascomycetes.

(131) Answer : (2)

Solution:

Genetic engineering involves the techniques to alter the chemistry of genetic material (DNA and RNA), to introduce these into host organism and thus change the phenotype of the host organism.

There are three basic steps involved in genetically modifying an organism-

(i) Identification of DNA with desirable genes

(ii) Introduction of the identified DNA into the host

(iii) Maintenance of introduced DNA in the host and transfer of the DNA to its progeny

(132) Answer : (1)

Solution:

A special coronary system of blood vessels is present in our body exclusively for the circulation of blood to and from the cardiac musculature. Coronary artery carries oxygenated blood to the heart muscles.

(133) Answer : (2)

Solution:

Based on the number of axon(s) and dendrite(s), the neurons are divided into three types, i.e.,

(A) Multipolar (with one axon and two or more dendrites; found in the cerebral cortex)

(B) Bipolar (with one axon and one dendrite, found in the retina of eye)

(C) Unipolar (cell body with one axon only; found usually in the embryonic stage)

(134) Answer : (1)

Solution:

Carpals (wrist bones) in one forelimb – 8

Cranial bones – 8

Facial bones – 14

Tarsals (ankle bones) in one hindlimb – 7

False ribs – 4

Cervical vertebrae – 7

(135) Answer : (4)

Solution:

Inulin is a homopolymer of fructose. Insulin is a protein and proteins are heteropolymers.

(136) Answer : (4)

Solution:

After performing agarose gel electrophoresis, the separated DNA fragments can be visualised only after staining the DNA with a compound known as ethidium bromide followed by exposure to UV radiation as one cannot see pure DNA fragments in the visible light and without staining.

(137) Answer : (3)

Solution:

During the extension step of PCR, nucleotides are required, not ATP.

Two types of primers are required during the annealing step, they are forward and reverse primers.

During the denaturation step of PCR, hydrogen bonds are broken to generate single stranded template DNA from the source dsDNA.

(138) Answer : (1)

Solution:

The rise in the stimulus-induced permeability to Na^+ is extremely short lived. It is quickly followed by a rise in permeability for K^+ . Within a fraction of a second, K^+ diffuses outside the membrane and restores the resting potential of the membrane at the site of excitation and the fibre becomes responsive to further stimulation.

(139) Answer : (4)

Solution:

Transformants can either be non-recombinant transformants or recombinant transformants.

Gene for β -galactosidase is a colour producing selectable marker.

Restriction enzyme does not act on the bacteria itself; it protects the bacteria from bacteriophages.

(140) Answer : (1)

Solution:

A classic case in medicine, is that of Mary Mallon nicknamed Typhoid Mary. She was a cook by profession and was a typhoid carrier who continued to spread typhoid for several years through the food she prepared.

(141) Answer : (4)

Solution:

Catecholamines are emergency hormones, also called hormones of fight or flight. They increase alertness, pupillary dilation, piloerection (raising of hair), sweating., etc.

They increase strength of heart contraction and rate of respiration.

They stimulate breakdown of glycogen resulting in increased concentration of glucose in blood. In addition, they also stimulate breakdown of lipids and proteins.

(142) Answer : (4)

Solution:

Graft rejection is mediated mainly by T-lymphocytes.

(143) Answer : (4)

Solution:

Since the nerve fibres do not innervate all cells of the body and cellular functions need to be continuously regularized; a special kind of coordination and integration has to be provided. This function is carried out by hormones. The neural system and the endocrine system jointly coordinate and regulate the physiological functions in the body.

(144) Answer : (1)

Solution:

Patient becomes so immune-deficient in case of HIV infection that he/she is unable to protect himself/herself against minor infections which otherwise could have been overcome.

Treatment with anti-retroviral drugs is only partially effective. They can prolong the life of patient but cannot prevent death.

(145) Answer : (1)

Solution:

Enzymes are divided into 6 classes each with 4-13 subclasses and named accordingly by a four-digit number

Class I – Dehydrogenases

Class II – Transferases

Class III – Hydrolases

Class IV – Lyases

Class V – Isomerases

Class VI – Ligases

(146) Answer : (4)

Solution:

Progestasert and LNG-20 are hormonal IUDs.
Multiload-375 is a copper releasing IUD.

(147) Answer : (2)

Solution:

Ancylostoma (Aschelminth) is circular in cross-section and possesses pseudocoelom. They have a complete alimentary canal and organ-system level of body organisation. It infects intestine of humans.

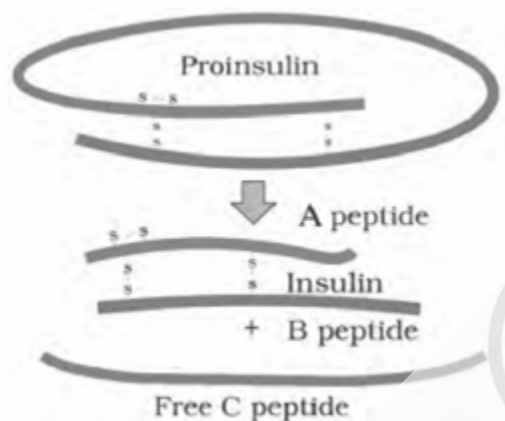
(148) Answer : (3)

Solution:

Organism 'X' is an echinoderm. The most distinctive feature of echinoderms is the presence of water vascular system which helps in locomotion, capture and transport of food and respiration. An excretory system is absent. Development is indirect with free-swimming larvae.

(149) Answer : (3)

Solution:



(150) Answer : (2)

Solution:

Fossils are generally found in sedimentary rocks.

(151) Answer : (3)

Solution:

Fossils of *Homo erectus* were discovered in Java in 1891. They had a large cranial capacity of around 900 cc.

(152) Answer : (3)

Solution:

Except for crocodiles, all other reptiles have three-chambered heart. Crocodiles have four-chambered heart. Rest of the features are common to most of the reptiles.

(153) Answer : (4)

Solution:

PCT is lined by simple cuboidal brush-border epithelium. Nearly all of the essential nutrients and 70-80 per cent of electrolytes and water are reabsorbed by this segment.

(154) Answer : (3)

Solution:

Gene flow, genetic drift, mutations, selective mating, natural selection and genetic recombination are the factors which affect Hardy-Weinberg equilibrium.

(155) Answer : (3)

Solution:

The mechanism of how non-cellular aggregates of giant macromolecules evolved into cells with eukaryotic cell is still majorly unknown.

(156) Answer : (4)

Solution:

Potato spindle tuber disease is caused by viroids and it was found to be free RNA with low molecular weight.

(157) Answer : (3)

Solution:

In racemose inflorescence, the flowers are borne in an acropetal succession. *i.e.* younger flowers are present towards the apex and the older flowers are present at the base.

(158) Answer : (3)

Solution:

Solanaceae family is commonly called potato family.

Flowers	–	Bisexual, actinomorphic
Inflorescence	–	Solitary, axillary or cymose
Corolla	–	Petals five, gamopetalous, valvate aestivation.

(159) Answer : (4)

Solution:

When stamens are attached to the perianth then the condition is called epiphyllous.

(160) Answer : (3)

Solution:

Typhoid is caused by bacteria.

Viruses are obligate parasites and are inert outside their specific host cell.

(161) Answer : (3)

Solution:

Both macro and micro elements are required by the plants for the synthesis of protoplasm and act as source of energy.

(162) Answer : (2)

Solution:

Interfascicular cambium is formed as a result of dedifferentiation.

(163) Answer : (2)

Solution:

Sporogenous tissue occupies the centre of each microsporangium.

(164) Answer : (2)

Solution:

Germ pores are the prominent apertures on the exine of a pollen grain which allow the pollen tube to emerge during germination.

(165) Answer : (1)

Solution:

Parthenium came into India as a contaminant with imported wheat and has become ubiquitous in occurrence and causes pollen allergy.

(166) Answer : (2)

Solution:

If starch grain size of pea is considered as the phenotype, the alleles show incomplete dominance.

(167) Answer : (2)

Solution:

Human skin colour is an example of polygenic trait which is not so distinct in occurrence and is spread across a gradient. It is darkest when the genotype has all the dominant alleles.

(168) Answer : (2)

Solution:

Initial clue of chromosomal mechanism of sex determination can be tracked back to the experiments carried out in insects. Large number of insects show XO type of sex determination. Drone honeybees are produced as a result of parthenogenesis.

(169) Answer : (3)

Solution:

According to Mendel's law of segregation, characters segregate during the formation of gametes.

(170) Answer : (1)

Solution:

Aleuroplasts are a type of leucoplasts, which are involved in the protein storage.

(171) Answer : (2)

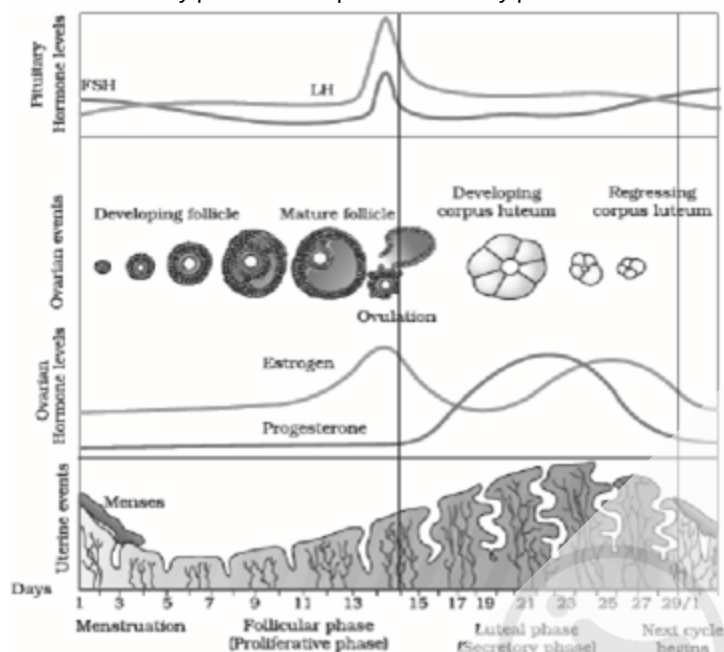
Solution:

ADH has constrictory effects on blood vessels. Renin converts angiotensinogen in blood to angiotensin I. ANF is a vasodilator.

(172) Answer : (3)**Solution:**

Menstrual cycle in human females consists of three phases.

1. Menstrual phase
2. Follicular phase or proliferative phase
3. Post-ovulatory phase/luteal phase/secretory phase

**(173) Answer :** (3)**Solution:**

Maintenance of regular menstrual cycle takes place in non-pregnant human females under control of GnRH from hypothalamus and gonadotropins from anterior pituitary.

(174) Answer : (3)**Solution:**

Crop (storage of food) is followed by proventriculus (grinding of food).

(175) Answer : (2)**Solution:**

By the end of 12 weeks (first trimester), most of major organ systems are formed.

(176) Answer : (1)**Solution:**

Vasa efferentia leave the testes and open into epididymis which continues as vas deferens. Urethra originates from urinary bladder, runs through penis and is responsible for transportation of both urine and semen. So, male urethra is commonly known as urinogenital duct.

(177) Answer : (3)**Solution:**

The correct route for passage of sperms in male frog is as follows

Testis → Vasa efferentia → Bidder's canal → Urinogenital duct → Cloaca

(178) Answer : (4)**Solution:**

Use of condoms has increased in recent years due to their additional benefit of protecting the users from contracting STIs and AIDS. Both the male and the female condoms are disposable, can be self-inserted and thereby give privacy to the users. They cannot be used as emergency contraceptive.

(179) Answer : (3)**Solution:**

Binding of oxygen with haemoglobin is primarily related to partial pressure of O_2 . Partial pressure of CO_2 , hydrogen ion concentration and temperature are the other factors which can interfere with this binding.

(180) Answer : (1)**Solution:**

Asthma is difficulty in breathing causing wheezing due to inflammation of bronchi and bronchioles. Emphysema is a chronic disorder in which alveolar walls are damaged due to which respiratory surface is decreased.

